

Gavin Xue

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EDUCATION

University of Waterloo

GPA: 3.9/4.0

Bachelor of Computer Science

Sep 2023 – Apr 2028

- Awarded **1 of 10** René Descartes National Scholarships (**\$20,000**) from the Faculty of Mathematics
- Relevant courses: Data Structures and Algorithms, Operating Systems, Databases, Concurrency, Artificial Intelligence

EXPERIENCE

Shopify

Toronto, ON

Software Engineering Intern

May 2026 – Aug 2026

- Designed a policy versioning framework for Shopify's internal trust platform using **Ruby on Rails**, letting teams publish new versions of YAML-defined compliance and fraud policies against an AST-validated schema
- Rolled 430K+ merchant records onto the latest policy version by backfilling **Bigtable** in incremental batches with Airflow-orchestrated **Python** scripts, with end-to-end observability in **Grafana** dashboards
- Prevented 1.4M+ record bulk backfills from overflowing the **Redis** job queue or delaying live **Kafka** event processing by adding queue-size throttling to the job pipeline

Software Engineering Intern

Sep 2025 – Dec 2025

- Collapsed 80+ forked components into shared base components across Shopify's checkout platform serving 2M+ merchants by unifying UI for 5 buyer identities in **React** and **TypeScript**
- Improved checkout maintainability for developers by eliminating 6K+ lines of forked code, designing a unified state layer that lifted identity logic into a shared **React** context

UW Blueprint

Waterloo, ON

Software Developer

May 2025 – Present

- Delivered reliable 25 MB file uploads across the club recruitment platform by designing a **GraphQL** multipart upload mutation that atomically rolls back **PostgreSQL** and **Firebase** writes across 3 failure paths
- Removed ~2h of manual reviewer-triage time by building an automated round-robin allocation service in **Express.js**
- Cut local dev setup from multiple days to <1 hour by switching to a monorepo and containerizing the services using **Docker**, standardizing dependency management across the **TypeScript** stack

Environment and Climate Change Canada

Toronto, ON

Software Developer Intern

Jan 2025 – Apr 2025

- Launched a climate forecasting product for 400+ researchers, using **React** and **D3.js** for interactive data visualizations
- Automated generation of 1,000+ contour plots and JSON outputs by building a data processing pipeline using **Python**, **pandas**, and **Matplotlib**
- Improved platform maintainability by refactoring legacy code into **MVC** architecture, reducing the codebase by 500+ lines

Develop for Good

Remote

Software Developer

Oct 2024 – Feb 2025

- Simplified daily admin tracking for an NGO's cat-neutering program by consolidating 5,000+ location records, previously scattered across spreadsheets and separate tools, into one geospatial platform built with **Next.js**
- Unified 3 fragmented data sources into a single **Supabase** database by designing an ETL pipeline that used **Python** and **GPT** to parse unstructured text across 50,000+ records

PROJECTS

notefuse 📄 | *Next.js, React, PostgreSQL, Redis, AWS S3, Pinecone, OpenAI*

- Built a RAG-based PDF analysis chatbot using **Next.js** and **LangChain**, implementing document chunking and vector similarity search across embedded text within a **Pinecone** vector database
- Reduced database queries using **Redis** caching layer to store user sessions and frequently accessed data

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Ruby, C++, C, Java, SQL, HTML, CSS, Bash

Frameworks: React, Node.js, Express.js, Ruby on Rails, PostgreSQL, GraphQL, pandas, Matplotlib

Infrastructure & Tools: GCP, AWS, Kafka, Redis, Docker, Kubernetes, Linux, Git